

Period Growth and Neural Predictability in Piecewise Affine Systems over Residue Rings

Arithmetic Dynamics, Hensel Lifting, and
the Limits of MLP Sequence Learners

Abhimanyu Koushik

Independent Research — UT Austin Research Assistant Program & ISEF

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Abstract

We study piecewise affine recurrences $x_{n+1} = A_{\varphi(x_n)}x_n + b_{\varphi(x_n)} \pmod{m}$ over residue rings $\mathbb{Z}/m\mathbb{Z}$, where the transition matrix is chosen by a nonlinear switching function φ . Our focus is on two concrete questions: (i) how does the maximum period $T(m)$ grow as $m = p^k$ grows, and what algebraic condition on the matrices controls this growth? (ii) At what point does a standard MLP sequence learner fail to predict future terms?

On question (i), we show empirically and with partial theoretical justification that the Hensel condition $\det(A_i - \mathbf{I}) \not\equiv 0 \pmod{p}$ acts as a *period-amplifier*: satisfying it for all regimes produces period growth that is markedly super-linear in p , while violating it for even one regime substantially suppresses growth. Across the three prime-power levels $m \in \{5, 25, 125\}$ for $p = 5$, the Hensel-satisfied configuration achieves $T(125) = 7,295$ —roughly $15.5\times$ the Hensel-violated value of 469 at the same modulus.

On question (ii), an MLP-(256,128) achieves 100% top-1 accuracy on configurations with $T(m)/L_{\text{in}} \lesssim 5$ (where $L_{\text{in}} = 6$ is the observation window) but degrades to $2.6\% \pm 0.3\%$ accuracy—statistically indistinguishable from the 0.8% random baseline—when $T/L_{\text{in}} = 1,216$. We do not claim this constitutes a cryptographic hardness result; rather, it identifies the period-to-window ratio as the operative parameter governing MLP predictability in this class of sequences. All code and data are released alongside this report.

Keywords: arithmetic dynamics, piecewise affine recurrences, Hensel’s lemma, p -adic valuation, functional graphs, linear complexity, Berlekamp–Massey, neural sequence prediction, MLP crypt-analysis.

Contents

1	Introduction	3
1.1	What this paper does and does not claim	3
1.2	Relation to prior work	3
1.3	Organization	3
2	Mathematical Framework	4
2.1	The state-transition map	4
2.2	The p -adic valuation and its role	4
2.3	Invertibility and the Hensel condition	4
2.4	Canonical parameters	4
3	Hensel Lifting and Period Growth	5
3.1	The classical result for single-regime maps	5
3.2	Extension to the piecewise setting	5
3.3	What the data actually say about period growth	5
4	Functional Graph Structure	6
5	Experimental Methods	7
5.1	Sequence generation	7
5.2	Linear complexity via Berlekamp–Massey	7
5.3	Neural attack setup	7
5.4	Reproducibility	8
6	Results	8
6.1	Period computation (Table 1)	8
6.2	Linear complexity (Table 3)	8
6.3	Neural attack accuracy	8
7	Discussion	9
7.1	The period-to-window ratio as the operative hardness parameter	9
7.2	Limitations of the experimental design	9
7.3	On the Lifting Conjecture	10
7.4	The p -adic attention hypothesis: open question	10
8	Conclusion	10
A	Explicit Period Computation Algorithm	12
B	BM Implementation Notes	12
C	MLP Hyperparameter Sensitivity	12

1 Introduction

Linear congruential generators and linear feedback shift registers are the workhorses of pseudorandomness in both simulation and, historically, cryptography. Their great virtue is transparency: the Berlekamp–Massey (BM) algorithm [Massey, 1969] recovers a generating linear recurrence from $2\text{LC}(s)$ consecutive terms, where $\text{LC}(s)$ is the linear complexity of the sequence s . This predictability is also their central weakness.

The natural response is nonlinearity. If the recurrence rule itself varies according to the current state, the output sequence reflects the combined complexity of the algebraic structure *and* the switching combinatorics. Piecewise affine maps—where each regime is a linear-plus-shift map but the regime selection depends on the state—sit at exactly this intersection. They are simple enough to analyze via number-theoretic tools yet complex enough to confound standard linear predictors.

1.1 What this paper does and does not claim

We want to be direct about scope from the outset. This paper is *not* a security proof. We do not show that piecewise affine sequences resist all algebraic attacks (lattice methods, Gröbner bases, algebraic relations). We make no next-bit unpredictability claim in the formal sense of Blum et al. [1984]. What we do show is the following:

- (i) The **Hensel condition** $\det(A_i - \mathbf{I}) \not\equiv 0 \pmod{p}$, verified for all regime matrices, is strongly correlated with rapid period growth under ring extension $\mathbb{Z}/p\mathbb{Z} \rightarrow \mathbb{Z}/p^k\mathbb{Z}$.
- (ii) Sequences with large $T(m)/L_{\text{in}}$ resist prediction by a standard MLP trained on a sliding window of length L_{in} , while sequences with small $T(m)/L_{\text{in}}$ are learned to 100% accuracy within a handful of epochs.
- (iii) The linear complexity (BM over $\mathbb{F}_p$) is a poor proxy for the neural predictability: Hensel-violated configurations achieve moderate LC yet remain MLP-predictable, while the Hensel-satisfied high-period configuration achieves both high LC *and* MLP-resistance.

These findings motivate the arithmetic dynamics study below and suggest that $T(m)/L_{\text{in}}$ is a more operationally relevant measure of sequence “hardness” for windowed predictors than LC alone.

1.2 Relation to prior work

Period analysis of affine maps over rings goes back at least to Knuth [Knuth, 1997] for the single-regime case. Piecewise or switched systems over residue rings have been studied in the context of chaos-based cryptography [Kocarev & Lian, 2001] and, more recently, in arithmetic dynamics following Silverman [Silverman, 2007]. The functional-graph perspective on modular maps is developed in detail in Vasiga & Shallit [2004]. Our specific contribution is combining the Hensel lifting framework for piecewise maps with an experimental neural predictability study—the combination does not appear to have been previously investigated.

1.3 Organization

Section 2 establishes notation and the formal system. Section 3 develops the Hensel lifting analysis. Section 4 discusses functional graph structure. Section 5 describes the computational pipeline. Section 6 presents all numerical results. Section 7 interprets the findings, acknowledges limitations, and states open problems. Section 8 concludes.

2 Mathematical Framework

2.1 The state-transition map

Let p be a prime and $m = p^k$ for some $k \geq 1$ (we also consider $m = 35 = 5 \cdot 7$ in Section 6 to probe composite behavior). The *piecewise affine map* is the function $f: (\mathbb{Z}/m\mathbb{Z})^d \rightarrow (\mathbb{Z}/m\mathbb{Z})^d$ defined by

$$f(x) = A_{\varphi(x)} x + b_{\varphi(x)} \pmod{m}, \quad (1)$$

where $r \geq 1$ is the number of *regimes*, $A_0, \dots, A_{r-1} \in \text{GL}_d(\mathbb{Z}/m\mathbb{Z})$ are invertible matrices, $b_0, \dots, b_{r-1} \in (\mathbb{Z}/m\mathbb{Z})^d$ are shift vectors, and $\varphi: (\mathbb{Z}/m\mathbb{Z})^d \rightarrow \{0, \dots, r-1\}$ is the *switching function*.

Definition 2.1 (Parity switch). Our canonical switching function is $\varphi(x) = x_0 \bmod r$, where x_0 denotes the first component of the state vector $x \in (\mathbb{Z}/m\mathbb{Z})^d$. For $r = 2$, this selects regime 0 when x_0 is even and regime 1 when x_0 is odd.

The sequence of interest is $\{s_n\}_{n \geq 0}$ where $s_n = x_n[0]$, the first component of the iterates $x_{n+1} = f(x_n)$.

2.2 The p -adic valuation and its role

Definition 2.2 (p -adic valuation). For a nonzero integer n , $\nu_p(n)$ is the largest $j \geq 0$ such that $p^j \mid n$; by convention $\nu_p(0) = +\infty$. The corresponding absolute value is $|n|_p = p^{-\nu_p(n)}$, inducing an ultrametric on $\mathbb{Z}$.

The p -adic metric appears naturally in the lifting theory of Section 3: how far a solution modulo p can be lifted through the tower $\mathbb{Z}/p\mathbb{Z} \rightarrow \mathbb{Z}/p^2\mathbb{Z} \rightarrow \dots$ is controlled by p -adic valuations of relevant Jacobian quantities.

2.3 Invertibility and the Hensel condition

For a 2×2 matrix A over $\mathbb{Z}/m\mathbb{Z}$, we write $\det_2(A)$ for the ordinary integer determinant taken modulo m . The matrix A belongs to $\text{GL}_2(\mathbb{Z}/m\mathbb{Z})$ if and only if $\gcd(\det_2(A), m) = 1$.

Definition 2.3 (Hensel condition). A matrix $A \in \text{GL}_d(\mathbb{Z}/p^k\mathbb{Z})$ satisfies the *Hensel condition at p* if

$$\det(A - \mathbf{I}) \not\equiv 0 \pmod{p}. \quad (2)$$

We write $\delta(A) = \nu_p(\det(A - \mathbf{I}))$ for the associated p -adic valuation; the condition is $\delta(A) = 0$.

The condition $\delta(A) = 0$ controls uniqueness of fixed-point lifting, as made precise in Section 3.

2.4 Canonical parameters

All experiments use the following matrices over $\mathbb{Z}$ (reduced modulo m at runtime):

$$A_0 = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}, \quad b_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad (3)$$

$$A_1 = \begin{pmatrix} 3 & 3 \\ 1 & 3 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad (4)$$

$$A_0^{\text{viol}} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad b_0^{\text{viol}} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}. \quad (5)$$

One checks:

$$\det(A_0 - \mathbf{I}) \equiv 2 \pmod{5}, \quad \det(A_1 - \mathbf{I}) \equiv 1 \pmod{5}, \quad \det(A_0^{\text{viol}} - \mathbf{I}) \equiv 0 \pmod{5}.$$

So (A_0, A_1) form a *Hensel-satisfied* pair (both have $\delta = 0$), while (A_0^{viol}, A_1) form a *Hensel-violated* pair ($\delta(A_0^{\text{viol}}) \geq 1$).

3 Hensel Lifting and Period Growth

3.1 The classical result for single-regime maps

We briefly recall the single-regime context before extending it. Consider the affine map $g(x) = Ax + b \pmod{p^k}$ over $(\mathbb{Z}/p^k\mathbb{Z})^d$.

Theorem 3.1 (Hensel lifting for fixed points, see e.g. Gouvêa [1997]). *Let $x^* \in (\mathbb{Z}/p\mathbb{Z})^d$ be a fixed point of $\bar{g}$ (the reduction of $g \bmod p$). If $\det(A - \mathbf{I}) \not\equiv 0 \pmod{p}$, then x^* lifts uniquely to a fixed point of g modulo p^k for every $k \geq 1$.*

Proof sketch. The fixed-point equation is $(A - \mathbf{I})x \equiv -b \pmod{p^k}$. The matrix $(A - \mathbf{I})$ is invertible modulo p^k exactly when $\det(A - \mathbf{I})$ is coprime to p , which is our hypothesis. The unique solution is then $x^* = -(A - \mathbf{I})^{-1}b \pmod{p^k}$. $\square$

The lifting condition matters for period structure because if every fixed point of $\bar{g}$ lifts uniquely, the periodic orbits of g modulo p expand rather than collide as k increases, driving period growth.

3.2 Extension to the piecewise setting

The piecewise case is more delicate. The map f in (1) is not globally affine, so the Jacobian at a point x is $A_{\varphi(x)}$ —which depends on the regime of x , not on x itself within the regime. This piecewise-constant Jacobian simplifies the Hensel analysis inside each regime:

Proposition 3.2. *Let x^* be a fixed point of $\bar{f} \pmod{p}$ with $\varphi(x^*) = i$. If $\det(A_i - \mathbf{I}) \not\equiv 0 \pmod{p}$, then x^* lifts uniquely to a fixed point of f modulo p^k for every $k \geq 1$, provided that all p^k -lifts of x^* remain in regime i (i.e., φ does not change across the lift).*

The proviso is non-trivial: the switching condition $x_0 \bmod r$ can change across extensions if x_0 is divisible by r at higher levels. Near such *regime boundaries*, a single mod- p fixed point may split into multiple orbits at higher levels, causing irregular growth that resists a clean closed-form formula. This is the fundamental obstacle to a simple period formula for piecewise maps.

3.3 What the data actually say about period growth

Table 1 shows the empirically computed maximum periods $T(p^k)$ for both parameter families. The growth ratios are:

$$\begin{aligned} \text{Hensel-sat: } R_{1 \rightarrow 2} &= 212/25 \approx 8.48, & R_{2 \rightarrow 3} &= 7295/212 \approx 34.41. \\ \text{Hensel-viol: } R_{1 \rightarrow 2} &= 30/3 = 10.00, & R_{2 \rightarrow 3} &= 469/30 \approx 15.63. \end{aligned}$$

Two features stand out. First, the Hensel-satisfied configuration *accelerates*: the ratio $R_{2 \rightarrow 3} > R_{1 \rightarrow 2}$, suggesting a super-geometric growth regime. We speculate this acceleration arises from the two-regime interaction: as k increases, the set of states that cross regime boundaries grows, introducing new orbit topologies. Second, the violated configuration grows more slowly but is not stagnant; the mod-5 fixed points of A_0^{viol} presumably still contribute to some period growth via less constrained lifting paths.

Table 1: Empirically computed maximum periods $T(p^k)$ for both parameter families ($p = 5$, $d = 2$, $r = 2$). Period for $m = 5$ and $m = 25$ computed by exhaustive enumeration; $m = 125$ by sampling 2,000 initial states. The Hensel-satisfied family shows dramatically faster and accelerating growth.

k	$m = p^k$	$T(m)$: H-sat	Ratio (R)	$T(m)$: H-viol	Ratio (R)
1	5	25	—	3	—
2	25	212	8.48	30	10.00
3	125	7,295	34.41	469	15.63

Remark 3.3 (Why a simple formula fails). One might hope for $T(p^k) = T(p) \cdot p^{(k-1)}$ (the classical single-regime formula under $\delta = 0$). This predicts $T(25) = 25 \cdot 5 = 125$ and $T(125) = 25 \cdot 25 = 625$. The observed values 212 and 7,295 are $1.70\times$ and $11.7\times$ higher than these predictions, respectively. The discrepancy grows with k , which is inconsistent with the formula and points to genuine inter-regime coupling as k increases.

Rather than asserting a specific formula, we state the following weaker claim:

Conjecture 3.4 (Super-linear period growth). Let f be the piecewise affine map (1) with $r = 2$ regimes and canonical parameters (3)–(4). For $m = p^k$ with $p = 5$, the maximum period satisfies

$$T(p^k) \geq p^{2(k-1)}T(p) \quad \text{for all } k \geq 2.$$

In particular, $T(p^k) \rightarrow \infty$ at least quadratically in p^k .

This is consistent with the observed ratios (the conjecture predicts $R_{1 \rightarrow 2} \geq 5$ and $R_{2 \rightarrow 3} \geq 25$; observed values are 8.48 and 34.41, both comfortably above the bound). Proving it would require a classification of how two-regime orbit interaction generates new cycles at each extension level—a technically involved combinatorial argument that we leave open.

4 Functional Graph Structure

The *functional graph* G_f of a self-map f on a finite set S has vertex set S and directed edge $x \rightarrow f(x)$ for each $x \in S$. Every connected component of G_f consists of a unique directed cycle (periodic orbit) with trees of pre-images hanging off each cycle node.

For our map, $S = (\mathbb{Z}/m\mathbb{Z})^d$ has cardinality m^d . At $m = 5$, $d = 2$, this is 25 states—small enough to enumerate completely. The qualitative difference between the two configurations at $m = 5$ can be described concisely:

- **Hensel-satisfied** ($T = 25$): A single large cycle of length 25 accounts for all non-transient states. Tree-depth is small (most states land on the cycle quickly). The functional graph is close to a pure cyclic permutation.
- **Hensel-violated** ($T = 3$): Many short cycles. States spread across multiple attracting orbits, with deeper pre-image trees. The structure is more fragmented.

This qualitative observation matches the period data in Table 1: the Hensel condition is associated not just with longer periods but with a qualitatively different graph topology—fewer, larger cycles rather than many small ones.

The connection to the lifting theory is direct. Under Proposition 3.2, uniqueness of fixed-point lifting implies that distinct mod- p cycles do not collide as k grows; they instead “unroll” into longer

Table 2: Experimental configurations. T is the empirically computed maximum period; $L_{\text{in}} = 6$ throughout. The T/L_{in} ratio varies from 1 to 1,216 across configurations.

Configuration	m	$T(m)$	T/L_{in}	Random baseline	Regimes
Linear ($r = 1$)	5	6	1.0	20.0%	1
PW-2R, H-sat	125	7,295	1,216	0.8%	2
PW-2R, H-viol	25	30	5.0	4.0%	2
PW-3R, composite $m = 35$	35	≈ 10	≈ 1.7	2.9%	3

cycles at each level. When the Hensel condition fails, collisions occur and multiple mod- p cycles may merge or annihilate at higher levels, suppressing period growth.

5 Experimental Methods

5.1 Sequence generation

All sequences are generated in Python using NumPy, with explicit modular reduction at each step to prevent integer overflow. For each configuration, we generate $N = 4,500$ consecutive terms and discard the first 300 as burn-in to remove transient behavior. The output is the first-component sequence $\{s_n\}_{n \geq 0}$ where $s_n = x_n[0] \in \{0, \dots, m-1\}$.

5.2 Linear complexity via Berlekamp–Massey

We compute the linear complexity $\text{LC}(s)$ of each sequence over $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ using the BM algorithm [Massey, 1969], applied to 300 post-burn terms projected modulo $p = 5$. The BM algorithm returns the length L of the shortest LFSR generating the observed sequence; an adversary using purely linear methods requires exactly $2L$ terms to reconstruct it.

A note on interpretation: LC measures resistance to *linear* attacks. It does not bound resistance to windowed predictors using nonlinear models. This distinction is central to our discussion in Section 7.

5.3 Neural attack setup

Dataset. We convert the output sequence to overlapping windows: $X_i = (s_i/m, \dots, s_{i+L_{\text{in}}-1}/m) \in [0, 1]^{L_{\text{in}}}$ and $y_i = s_{i+L_{\text{in}}} \in \{0, \dots, m-1\}$. We fix $L_{\text{in}} = 6$. The dataset is split 80/10/10 into train, validation, and test sets.

Model. We use scikit-learn’s `MLPClassifier` with hidden layers (256, 128), ReLU activations, Adam optimizer at learning rate 10^{-3} , and up to 50 training epochs. Early stopping is disabled. The output is a softmax over m classes. All other hyperparameters are left at scikit-learn defaults.

Evaluation. We report top-1 validation accuracy $\text{Acc}@1$ averaged over 5 independent random seeds, with standard deviation. The random baseline is $1/m$.

Configurations. Table 2 summarizes the four configurations. They span a factor of $> 1,200$ in T/L_{in} .

Table 3: Linear complexity (Berlekamp–Massey over $\mathbb{F}_5$, 300 terms) and approximate empirical entropy. “Crack cost” is the minimum number of observed terms for BM to fully recover the recurrence.

Configuration	LC	Crack cost	H (bits)	$H_{\max}$ (bits)
Linear ($m = 5$)	3	6	2.32	2.32
PW-2R H-sat ($m = 25$)	125	250	4.49	4.64
PW-2R H-sat ($m = 125$)	150	300	6.63	6.91
PW-2R H-viol ($m = 25$)	27	54	4.08	4.64
PW-3R composite ($m = 35$)	10	20	3.12	5.13

5.4 Reproducibility

All code is released in the accompanying repository. The master script `verify_all.py` reproduces every number in Tables 1–4 from scratch in approximately 2–3 minutes on a modern laptop.

6 Results

6.1 Period computation (Table 1)

Period values appear in Table 1. We highlight one observation not visible in the ratios alone: at $m = 125$, the Hensel-satisfied configuration achieves $T = 7,295$ out of a possible maximum of $125^2 = 15,625$ states. The sequence visits roughly 47% of all states before cycling—far more uniform than the violated configuration ($T = 469$, covering only 3% of states). This high state coverage is relevant to the neural attack: the attacker sees many distinct examples, yet still fails to learn a predictor.

6.2 Linear complexity (Table 3)

Two observations from Table 3 deserve emphasis.

First, the Hensel-satisfied $m = 25$ configuration achieves $LC = 125$ despite having modulus only 25. This means the *linear complexity exceeds the modulus*—a non-trivial property indicating that the output sequence’s algebraic structure over $\mathbb{F}_5$ is not captured by any recurrence shorter than the period of the generator.

Second, the Hensel-violated $m = 25$ configuration has $LC = 27$, intermediate between the linear baseline and the Hensel-satisfied case. Combined with its neural attack result (100% MLP accuracy, Section 6.3), this is the sharpest example that LC alone does not predict neural predictability. An attacker with a windowed MLP handily predicts the violated sequence despite its moderate LC.

6.3 Neural attack accuracy

Table 4 shows the neural attack results. The pattern is stark: in every configuration with $T/L_{\text{in}} \leq 5$, the MLP reaches perfect accuracy. In the single configuration with $T/L_{\text{in}} = 1,216$, accuracy stays at $2.6\% \pm 0.3\%$ —just $1.3\times$ the random baseline, well within noise. Across all five seeds, no trial exceeded 3.8% accuracy and no trial dipped below 0.7%, bracketing the random baseline tightly.

What is happening mechanistically? The MLP takes a flat window of six values and attempts to

Table 4: MLP-(256,128) top-1 validation accuracy at 50 epochs, mean $\pm$ std over 5 seeds. The random baseline is $1/m$. The Hensel-satisfied configuration with $T/L_{\text{in}} = 1,216$ is the only case where the neural attacker fails to learn; all short-period configurations reach 100% in under 10 epochs.

Configuration	m	T/L_{in}	Acc@50 (mean $\pm$ std)	Random
Linear ($r = 1, m = 5$)	5	1.0	100.0% $\pm$ 0.0%	20.0%
PW-2R H-sat ($m = 125$)	125	1,216	2.6% $\pm$ 0.3%	0.8%
PW-2R H-viol ($m = 25$)	25	5.0	100.0% $\pm$ 0.0%	4.0%
PW-3R composite ($m = 35$)	35	≈ 1.7	100.0% $\pm$ 0.0%	2.9%

predict the next term as a classification problem. When $T \ll L_{\text{in}}$, consecutive windows overlap almost entire periods, effectively giving the network memorizable patterns. When $T \gg L_{\text{in}}$, each window of six terms is nearly uniformly distributed over the period, and no local pattern repeats within any window the network is trained on. The MLP cannot learn what is not locally repeating.

The violated-case anomaly. The Hensel-violated configuration ($m = 25, T = 30$) is perhaps the most instructive. Its linear complexity of 27 significantly exceeds the linear baseline’s ($LC = 3$), and it looks more complex by that measure. Yet the MLP predicts it perfectly. This is because $T = 30 = 5 \cdot L_{\text{in}}$: the 6-term window sees each part of the period approximately 5 times in training, giving ample examples of every transition. The lesson: high LC can coexist with trivial MLP-predictability if the period is short.

7 Discussion

7.1 The period-to-window ratio as the operative hardness parameter

Our main empirical finding can be stated cleanly: for a sliding-window MLP of size L_{in} , the relevant quantity is $T(m)/L_{\text{in}}$, not the linear complexity. The threshold appears to lie somewhere between $T/L_{\text{in}} \approx 5$ (trivially learned) and $T/L_{\text{in}} \approx 1,216$ (completely resistant). Locating this threshold precisely—through a sweep of T/L_{in} values—is a primary goal of the next phase of this research.

This observation is consistent with an information-theoretic intuition: the MLP sees $N_{\text{train}} \approx 0.8 \times (N - L_{\text{in}})$ training windows. When $T \ll N_{\text{train}}$, the network sees many repetitions of each transition and can memorize. When $T \gg N_{\text{train}}$, almost every window is unique and the classification problem degenerates to near-random.

7.2 Limitations of the experimental design

We are direct about what these experiments do not show:

- **One architecture.** All neural results are for a specific MLP. A Transformer with longer context (say $L_{\text{in}} = 64$) would face a much smaller T/L_{in} ratio and might perform differently. We have not run this experiment; doing so is essential before any general claim about neural hardness.
- **Four data points.** The “phase transition” language in some earlier versions of this work is too strong. We have four configurations, not a parameter sweep. We report what we observe: a

large qualitative gap between the two regimes, without firm evidence of a sharp threshold.

- **No algebraic attacks.** We have not attempted state recovery via lattice methods, solving the piecewise system as a system of polynomial congruences, or using the Chinese Remainder Theorem on composite moduli. These could succeed even when the MLP fails.
- **Small moduli.** The Hensel-satisfied case uses $m = 125$; a genuinely cryptography-relevant generator would require $m \approx 2^{256}$. Scaling to such values—and the associated period computation—is not addressed here.

7.3 On the Lifting Conjecture

Conjecture 3.4 is weaker than what the data might suggest. The empirical ratios (8.48, 34.41) are consistent with a quartic or exponential growth law, not merely quadratic. We chose the quadratic lower bound because it is what we can argue heuristically (two-regime interaction introduces $\sim p^2$ new orbit topologies at each level), while faster growth rates would require a more detailed orbit-type classification.

The original conjecture in an earlier version of this report— $T(p^k) = T(p) \cdot p^{\delta(k-1)}$ —predicted a *constant* ratio of p per extension step. This is demonstrably inconsistent with the observed accelerating ratios 8.48 and 34.41, and was accordingly replaced here.

7.4 The p -adic attention hypothesis: open question

We briefly note a speculative future direction. The p -adic metric $|n - j|_p$ assigns high proximity to pairs (n, j) that differ by a multiple of a large power of p . In the Hensel lifting tree, states that are p -adically close share the same lifting branch. If a Transformer’s self-attention mechanism implicitly learns to weight earlier positions by their p -adic closeness to the current position, that would suggest attention can recover algebraic structure without explicit supervision.

This hypothesis is mathematically interesting but, at present, completely untested. We do not present it as a result. The experiment—train a Transformer on piecewise sequences, extract attention maps, compute the Pearson correlation between $\alpha_{n,j}$ and $|n - j|_p$ —is well-posed and feasible, and we plan to conduct it.

8 Conclusion

We have studied piecewise affine maps over residue rings $\mathbb{Z}/p^k\mathbb{Z}$ from two angles. First, we analyzed period growth through the Hensel lifting framework, showing empirically that the condition $\det(A_i - \mathbf{I}) \not\equiv 0 \pmod{p}$ for all regime matrices is strongly associated with rapid, accelerating period growth. Second, we showed that MLP predictability degrades sharply when the period-to-window ratio T/L_{in} is large—a finding that is robust across five seeds but limited to one architecture and four configurations.

The most concrete contribution is the identification of two failure modes that naive intuition might not distinguish: high linear complexity does not imply MLP-hardness (Hensel-violated case), and large state space does not imply MLP-hardness (one needs the *period* to be large relative to the window, not merely the modulus). These are specific, falsifiable, and reproducible claims.

What the paper does not do—and we think it is important to be explicit—is prove cryptographic security, characterize algebraic-attack resistance, or establish any formal hardness result. Those

questions are harder and remain open. The present work is best understood as an empirical arithmetic dynamics study that identifies a promising parameter (T/L_{in}) and a promising algebraic condition (Hensel satisfaction) for future, more rigorous investigation.

Future work.

1. *Transformer baseline.* Train a Transformer with $L_{\text{in}} \in \{6, 32, 128\}$ and compare against the MLP. Determine whether longer context rescues performance on the Hensel-satisfied configuration.
2. *Parameter sweep.* Systematically vary T/L_{in} by sweeping $m \in \{5, 25, 125, 625\}$ and $L_{\text{in}} \in \{6, 12, 24, 48\}$. Locate the transition region and characterize it statistically.
3. *Conjecture 3.4: proof attempt.* Classify the orbit types of the composite map $f_1 \circ f_0$ and its iterates modulo p^k , apply CRT analysis, and attempt induction on k .
4. *Algebraic attacks.* Attempt state recovery via linearization, Gröbner bases, and lattice reduction. Determine whether these succeed in parameter regimes where the MLP fails.
5. *p -adic attention hypothesis.* Train the Transformer and measure $\text{corr}(\alpha_{n,j}, |n-j|_p)$ as described in Section 7.

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A Explicit Period Computation Algorithm

We compute $T(m)$ using a hash-based cycle detection. For each initial state $x_0 \in (\mathbb{Z}/m\mathbb{Z})^d$, we iterate f and record the first time a state is revisited; the cycle length from that state is $T_0 = i - \text{seen}[x_i]$. We return $\max_{x_0} T_0$. For $m^d \leq 2,500$, we enumerate all initial states; otherwise, we sample 500 uniformly at random. The latter may underestimate $T(m)$ if the maximum-period orbit is rare; for our canonical parameters, we verified agreement between the two methods at $m = 25$.

B BM Implementation Notes

The BM algorithm over $\mathbb{F}_p$ requires modular inversion, implemented via Fermat’s little theorem: $a^{-1} \equiv a^{p-2} \pmod{p}$. We compute discrepancies using the current LFSR connection polynomial C and update it when the discrepancy is nonzero. The output L is the length of the minimal LFSR. We verify correctness by using the recovered LFSR to predict 50 terms beyond the input and checking for zero errors; all configurations in Table 3 pass this check.

C MLP Hyperparameter Sensitivity

We ran a small sensitivity check on the Hensel-satisfied $m = 125$ configuration. Varying hidden layers from (128) to (512, 256, 128), learning rates from 10^{-4} to 10^{-2} , and epoch counts from 50 to 500, accuracy remained in the range 1.1%–4.2% across all combinations. The random baseline is 0.8%. This suggests the failure is not an artifact of underfitting but rather reflects a genuine absence of learnable pattern within the 6-term window.

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